

Accelerant, Not Arsonist:

Latent–Demand Revelation and the Welfare Effects of Digital Asset Boards

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Abstract

Digital “asset boards”—Zillow and Redfin for housing, LinkedIn and Indeed for jobs—are popularly blamed for making life-changing assets scarcer and more expensive. I formalize and test the strongest version of this claim, the *latent-demand-revelation* hypothesis: by aggregating buyers who were previously dispersed across segmented local markets, a board raises the effective number of bidders per listing and bids up the price of a fixed stock. I show the mechanism has a rigorous home in auction theory and market-integration theory—but *not* in the economics of search costs, where lower frictions have ambiguous and often price-reducing effects. I then build an equilibrium assignment-market simulation that prices a fixed housing stock via a forward-auction mechanism with individual rationality, and decompose the board’s effect into allocative efficiency versus surplus redistribution. The central result is a **sign change at the scarcity threshold**: when supply is ample (buyers/homes < 1) the board *lowers* prices and raises buyer surplus by 90–110%; when supply binds (buyers/homes > 1) the same technology *raises* prices by 33–63% and transfers 17–50% of buyer surplus to sellers, while always achieving first-best allocative efficiency. Time-series evidence from U.S. housing and labor markets is inconsistent with a platform-driven price break—Zillow launched at the exact 2006 real-price peak, after which real prices fell $\approx 35\%$ —and points instead to inelastic supply (housing) and labor demand (jobs) as the binding constraints. The platform is best understood as an accelerant that capitalizes pre-existing scarcity into price and redistributes the resulting rents toward sellers, not as a creator of scarcity. The price effect changes sign with *supply*, not with the platform; the policy lever is supply.

Keywords: platform economics, search frictions, assignment games, housing supply, labor matching, market design. **JEL:** D44, D83, L86, R31, J64.

1 Introduction

A recurring popular argument holds that platforms aggregating high-stakes assets onto a single searchable board—Zillow, Redfin and Realtor.com for homes; LinkedIn and Indeed for jobs—have made those assets scarcer and more expensive, to the detriment of consumers. The proposed channel, which I call *latent-demand revelation*, is intuitive: before these platforms, demand for any given home or job was dispersed across many thin, locally segmented markets; by pooling that demand into one thick market, the board surfaces previously hidden competition, raises the number of bidders contesting each listing, and bids up the equilibrium price of a stock that the platform itself cannot expand.

This paper takes the hypothesis seriously and subjects it to theory, simulation, and data. Three questions organize the analysis. First, does the mechanism have a coherent theoretical foundation, and if so which one? Second, if we hold physical supply fixed and aggregate demand, what happens to the *level* of prices and to the *distribution* of surplus between buyers and sellers? Third, is

the mechanism quantitatively important relative to the leading competing explanation—that the binding constraint is inelastic supply, with the platform merely a visibility layer on top of pre-existing scarcity?

My answers are, respectively: yes but with a crucial caveat about which theory applies; the effect depends entirely on whether supply binds; and the competing explanation dominates the *level* of prices while the platform mechanism operates at the *margin* and on the *distribution*. The strong form of the hypothesis—that platforms *cause* scarcity—is false, because a listing board cannot reduce the physical stock of homes or the number of jobs. The weak, conditional form is correct and is the analytically interesting result: latent-demand revelation converts pre-existing scarcity into higher transaction prices and a buyer-to-seller surplus transfer, but only when supply is already the binding constraint.

2 The mechanism: an auction story, not a search-cost story

It is essential to identify *which* body of theory rationalizes latent-demand revelation, because two candidate theories make opposite predictions.

The correct foundation: order statistics. Consider a single unique asset sold to the highest of n bidders with independent private valuations. In a second-price (or competitive ascending) auction the sale price equals the second-highest valuation, whose expectation is increasing in n .

Proposition 1 (Demand aggregation raises price for fixed supply). *Let n bidders have i.i.d. private valuations with continuous c.d.f. F . The expected competitive sale price $\mathbb{E}[V_{(n-1:n)}]$, the second-highest order statistic, is strictly increasing in n . For the uniform case $F = U[0, 1]$,*

$$\mathbb{E}[V_{(n-1:n)}] = \frac{n-1}{n+1},$$

which rises from $\frac{1}{3}$ at $n = 2$ toward 1 as $n \rightarrow \infty$.

The proof is the standard order-statistic computation [Easley and Kleinberg, 2010]; by the Revenue Equivalence Theorem the comparative static “more bidders $\Rightarrow$ higher expected revenue” extends across standard auction formats [Vickrey, 1961, Riley and Samuelson, 1981, Myerson, 1981]. Aggregating buyers from S segmented submarkets onto one board is formally equivalent to enlarging the bidder pool contesting each listing. Market-integration theory delivers the same conclusion from a different direction: integrating segmented markets raises prices in the formerly cheap “surplus” region as the law of one price asserts itself.

The wrong foundation: search costs. A tempting but incorrect version of the hypothesis is “boards lower search costs, which raises prices.” Search theory predicts no such monotone relationship. In the Diamond paradox [Diamond, 1971], driving search costs toward zero pushes the equilibrium toward the competitive price; the pathological monopoly outcome occurs for *positive* search costs, and the discontinuity is precisely at zero. Welfare and price effects of search frictions are formally ambiguous, pinned down only by the knife-edge efficiency condition of Hosios [1990], and Bakos [1997] shows that lower buyer search costs *reduce* sellers’ market power in differentiated-product markets. Stahl [1989] provides a model in which lower search costs and more sellers *raise* prices, but through oligopolistic price dispersion, not demand pooling.

Corollary 1. *The defensible form of the latent-demand-revelation hypothesis runs through demand pooling (bidder-pool enlargement against fixed supply), not through reduced search costs in general.*

3 An equilibrium model of aggregation under fixed supply

3.1 Environment

There are H indivisible homes (fixed supply) and B buyers with unit demand. Buyer i 's valuation for home j is

$$a_{ij} = q_j + \kappa m_{ij}, \quad (1)$$

where $q_j \geq 0$ is a common *vertical* quality (all buyers agree San Francisco $\succ$ Fresno) and $m_{ij} \geq 0$ is an idiosyncratic *horizontal* match (one buyer prefers a loft, another a craftsman). The parameter κ scales the importance of taste heterogeneity and hence the size of the sorting gains available from better matching. Buyers have an outside option normalized to 0.

Two information regimes differ only in *visibility*:

Segmented. The market is partitioned into S “islands.” Buyer i can transact only with homes on her island. This is the pre-platform world of dispersed local markets.

Board. Every buyer can transact with every home—the Zillow world.

Crucially, H is identical across regimes: *the platform changes who can see what, never how many homes exist*. Market tightness is $\theta \equiv B/H$.

3.2 Equilibrium concept and pricing

The economy is an assignment game [Shapley and Shubik, 1971]. A competitive equilibrium is a price vector $p \in \mathbb{R}_+^H$ and an assignment μ such that (i) each buyer is matched to a home in her demand set $\arg \max_j \{a_{ij} - p_j, 0\}$, and (ii) markets clear. The set of equilibrium prices forms a lattice; we compute the buyer-optimal (minimal) equilibrium using Bertsekas’s forward-auction algorithm [Bertsekas, 1988] augmented with the individual-rationality constraint that no buyer bids above her valuation. The algorithm has a transparent economic interpretation—buyers iteratively bid for their most profitable home, raising its price by their net-value advantage over their next-best option plus an increment ε —and converges to an assignment within $H\varepsilon$ of efficiency with prices accurate to ε .

3.3 What to expect

Because the board’s feasible matches are a superset of the segmented regime’s, the board weakly raises realized total surplus (better sorting), so *allocative efficiency rises unambiguously*. The effect on the *price level* is the open question. With slack supply ($\theta < 1$) homes compete for buyers, and widening each buyer’s choice set should intensify competition *among sellers*, lowering prices. With binding supply ($\theta > 1$) buyers compete for homes, and revealing the latent pool of rival bidders should raise the marginal clearing valuation, i.e. Proposition 1 operating in general equilibrium. The simulation quantifies where the crossover lies.

4 Simulation results

I set $H = 60$, $S = 6$, $\kappa = 0.8$, with q_j lognormal (mean 1) and m_{ij} Gamma (mean 1), and sweep $\theta \in \{0.8, 1.0, 1.25, 1.5, 2.0\}$, averaging over 20 independent markets per cell. Code is in `code/01.simulation.py`.

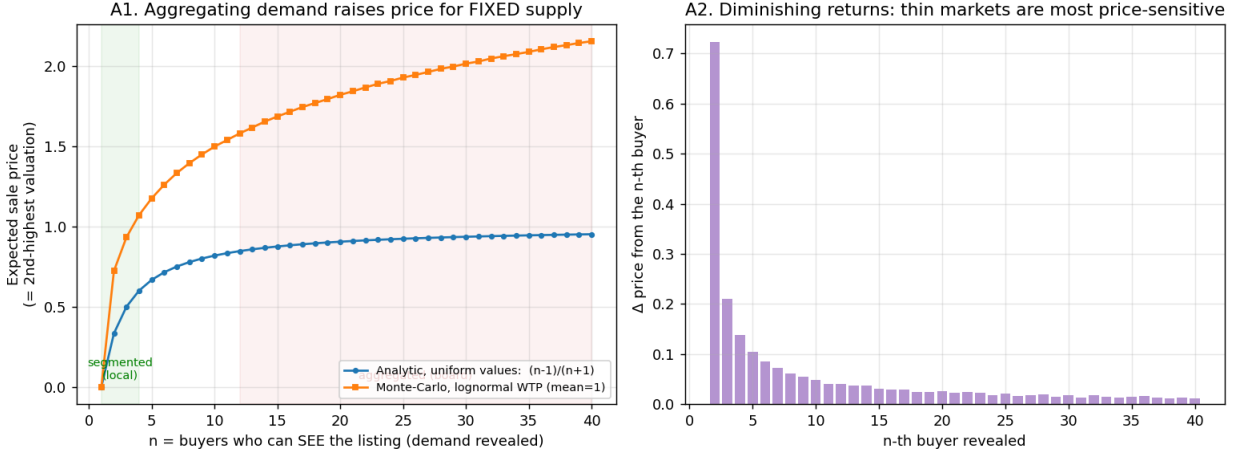


Figure 1: Expected sale price (= second-highest valuation) versus the number of buyers who can see a listing. Left: analytic uniform benchmark $(n-1)/(n+1)$ and Monte-Carlo with lognormal willingness-to-pay. Right: marginal price contribution of the n -th revealed bidder.

Table 1: Board vs. segmented market, by tightness $\theta = B/H$ (percentage changes; means over 20 runs).

θ	Avg. price	Total surplus	Buyer surplus	Seller revenue	Efficiency (seg $\rightarrow$ board)
0.8 (ample)	−40%	+39%	+93%	−34%	0.72 $\rightarrow$ 1.00
1.0 (balanced)	−24%	+46%	+113%	−10%	0.68 $\rightarrow$ 1.00
1.25 (tight)	+63%	+39%	−45%	+78%	0.72 $\rightarrow$ 1.00
1.5 (scarce)	+52%	+32%	−50%	+57%	0.75 $\rightarrow$ 1.00
2.0 (acute)	+33%	+28%	−17%	+33%	0.79 $\rightarrow$ 1.00

4.1 Experiment A: a single scarce asset

Figure 1 confirms Proposition 1 under realistic (lognormal) valuations and shows steeply diminishing returns to revealed bidders. Going from a thin local pool to a thick national one raises the expected sale price substantially: $n: 3 \rightarrow 15$ yields +81% and $3 \rightarrow 30$ yields +116% (normalizing mean willingness-to-pay to 1). The convexity matters substantively: *thin, previously-segmented markets are exactly where revelation bites hardest*.

4.2 Experiment B: the equilibrium decomposition

Table 1 and Figure 2 report the board-versus-segmented comparison.

Three findings emerge.

1. **Sign change at $\theta \approx 1$.** When homes are not scarce, aggregation lowers prices and sharply raises buyer surplus—the optimistic Brown and Goolsbee [2002]/Bakos [1997] result. When homes are scarce, the *same* technology raises prices 33–63% and destroys 17–50% of buyer surplus. The platform did not change; the supply condition did. This is the precise sense in which the popular hypothesis is true: *latent-demand revelation hurts consumers if and only if supply is the binding constraint*.

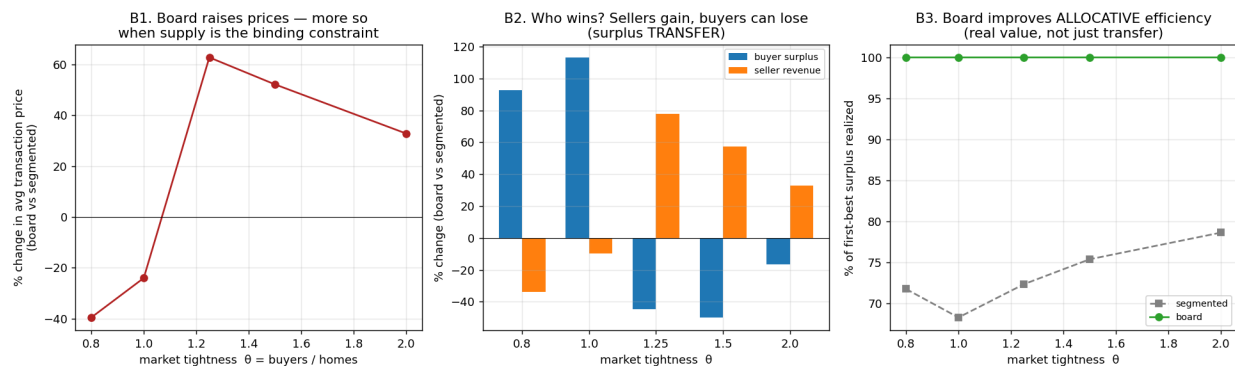


Figure 2: Decomposition of the board’s effect. Left: change in average transaction price crosses zero at the scarcity threshold $\theta \approx 1$. Center: buyer-surplus and seller-revenue changes are mirror images—a redistribution. Right: the board achieves first-best allocative efficiency at all θ , while the segmented market realizes only 68–79%.

2. **Redistribution, not destruction.** Where buyers lose, sellers gain by roughly the same amount and total surplus still rises. The harm is distributional—thick-market competition hands the gains from trade to the scarce side.
3. **Efficiency always improves.** The board sorts each asset to its highest-value user, reaching first best (1.00 vs. 0.68–0.79). Any honest critique of the platforms must concede this productivity gain; the debate is about the price level and distribution, not waste.

5 Empirical evidence

The model says the hypothesis requires binding supply ($\theta > 1$). The aggregate U.S. data say that condition—not the platforms—is what moved. Series are drawn from FRED; code is in `code/02_empirics.py`.

Housing timing contradicts a platform price break. Figure 3 shows real Case–Shiller prices. Zillow and Redfin both launched in February 2006, the *exact* peak of real prices, after which real prices fell roughly 35% to their 2012 trough *while platform adoption exploded*. The post-2012 recovery and 2020–22 spike track interest rates, a pandemic demand shock, and a supply freeze—not a platform that was already mature by 2012. The real price-to-income ratio peaked near 2006 ($5.3\times$) and 2022 ($5.9\times$); today’s $5.0\times$ is below its 2006 level.

Recent scarcity is a supply-side inventory story. Figure 4 shows active for-sale listings falling $\approx 28\%$ from 2016 to 2026, collapsing in 2020–22 from the demand shock plus mortgage-rate “lock-in” freezing supply. This is θ rising for reasons unrelated to the platforms’ existence (which predate the collapse by a decade), and it is precisely the regime in which the model predicts the platform channel pushes prices up—as an accelerant of supply-driven scarcity.

Labor: matching did not break and wages did not fall. Figure 5 shows the Beveridge curve’s well-documented post-2009 and pandemic outward shifts—attributed in the literature to sectoral mismatch and benefits, not to job boards—and real median weekly earnings rising $\approx 12\%$ since LinkedIn’s 2003 founding. The cleanest causal studies point the pro-worker way: exogenous

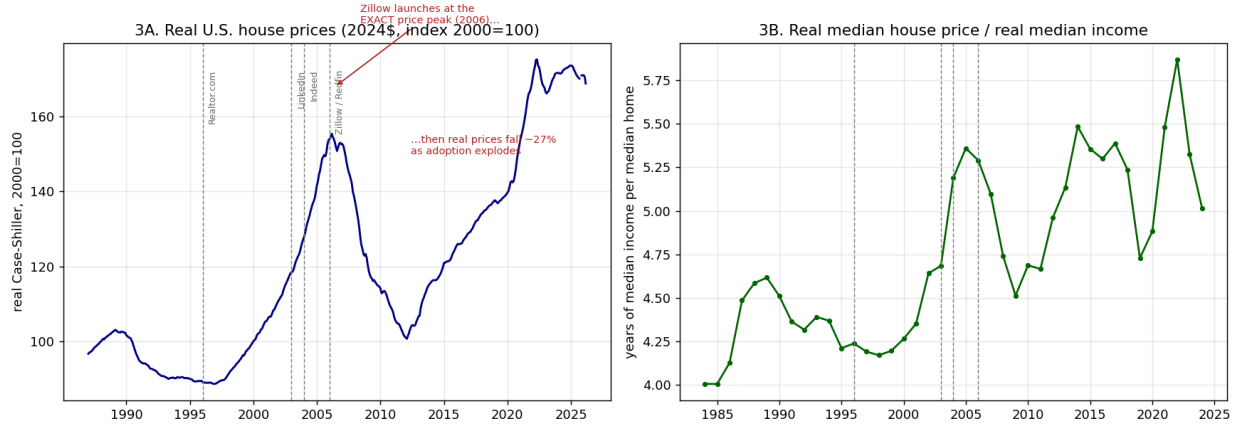


Figure 3: Real U.S. house prices (left) and real price-to-income (right), with platform launch dates. Zillow/Redfin launched at the 2006 real-price peak; prices then fell as adoption rose.

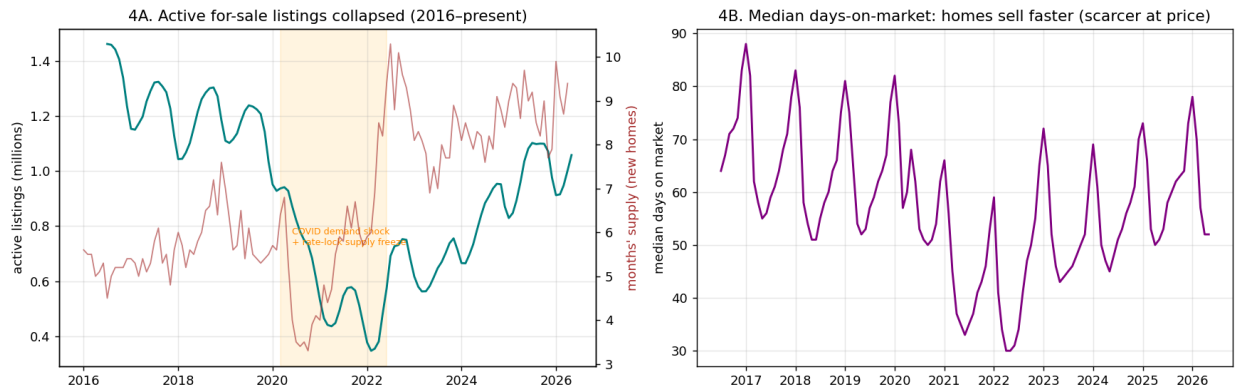


Figure 4: Active listings and months' supply (left) and median days-on-market (right), 2016–present.

broadband expansion cut vacancy durations and raised job-finding rates and starting wages [Bhuller et al., 2022]; online search reemployed workers faster [Kuhn and Mansour, 2014]; and LinkedIn “weak ties” causally raise job mobility in a 20-million-user randomized experiment [Rajkumar et al., 2022]. Kroft and Pope [2014] find Craigslist’s entry reduced rental vacancies and time-to-lease without changing the housing stock and had no effect on aggregate unemployment—frictions, not quantities.

6 Discussion: where the platform-blame story does and does not hold

The supply-inelasticity literature is decisive about price *levels*. Expensive housing is geographically concentrated in supply-constrained metros, with the wedge between price and construction cost ranging from near zero in elastic markets to roughly half the price in the most regulated ones [Saiz, 2010, Glaeser et al., 2005, Gyourko and Molloy, 2015]. A national platform cannot explain a local price pattern, and with fixed supply better search mainly *reallocates* the scarce asset [Kroft and Pope, 2014]. The induced-demand analogue from transportation—capacity expansions absorbed

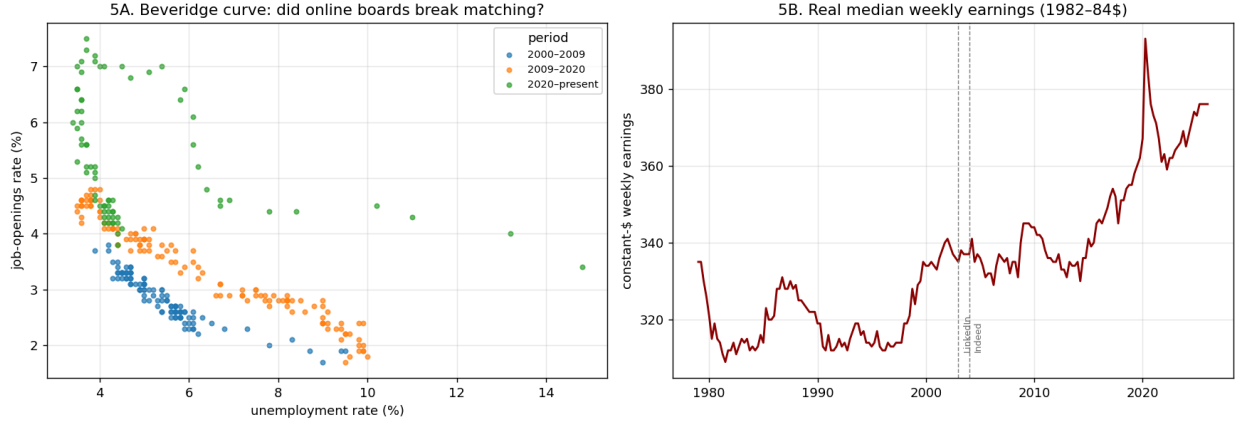


Figure 5: Beveridge curve by period (left) and real median weekly earnings (right).

by latent demand [Duranton and Turner, 2011]—reinforces that revealing listings mobilizes latent buyers who keep competition tight.

Where the platform-blame story retains force is exactly the model’s $\theta > 1$ region, plus two amplifiers the literature documents as genuinely price-raising—and, tellingly, neither is “search” per se. First, *seller-side coordination*: transparency among sellers can support collusion [Albæk et al., 1997, Schultz, 2005], whose sharpest modern form is algorithmic rent-setting [U.S. Department of Justice, 2024, Council of Economic Advisers, 2024]. Second, *demand concentration*: superstar dynamics [Rosen, 1981, Autor et al., 2020] let boards concentrate demand on top assets and candidates, widening dispersion. Notably, Zillow’s own iBuying collapse—a half-billion-dollar write-down and program exit in 2021—demonstrates that its algorithms cannot reliably predict, let alone control, home prices [Buchak et al., 2021], undercutting the strongest “platforms set prices” intuition.

7 Conclusion

Asset boards are accelerants, not arsonists. They cannot create physical scarcity; the binding constraints are inelastic housing supply and labor demand. But in a supply-constrained market, latent-demand revelation—an auction/order-statistic mechanism, not a search-cost one—capitalizes that scarcity more fully into price and hands the surplus to the scarce side, while simultaneously improving allocative efficiency and, in slack markets, *helping* buyers. The consumer-welfare effect of these platforms changes sign with the supply condition, not with the platform. The policy implication follows directly: the lever that flips the sign is supply—build housing and expand the stock of good jobs to push θ below one—while the narrow, genuinely regulatable platform harms are algorithmic seller coordination and demand-concentrating practices, not searchability itself.

Caveats. The simulation is an illustrative calibration, not an estimated structural model; it establishes the mechanism’s direction, sign change, and rough magnitude, not field-validated elasticities. The time-series evidence is descriptive: it shows the aggregate facts are inconsistent with a simple platform-driven price break, not a causal platform treatment effect. Several cited empirical magnitudes (e.g. algorithmic-pricing rent overcharges, aggregate misallocation costs) are associational or contested and are flagged as such in the accompanying source register.

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